

Portfolio health at a glance — churn exposure, revenue risk and contract breakdown

7,032

Total Customers

26.6%

Overall Churn Rate

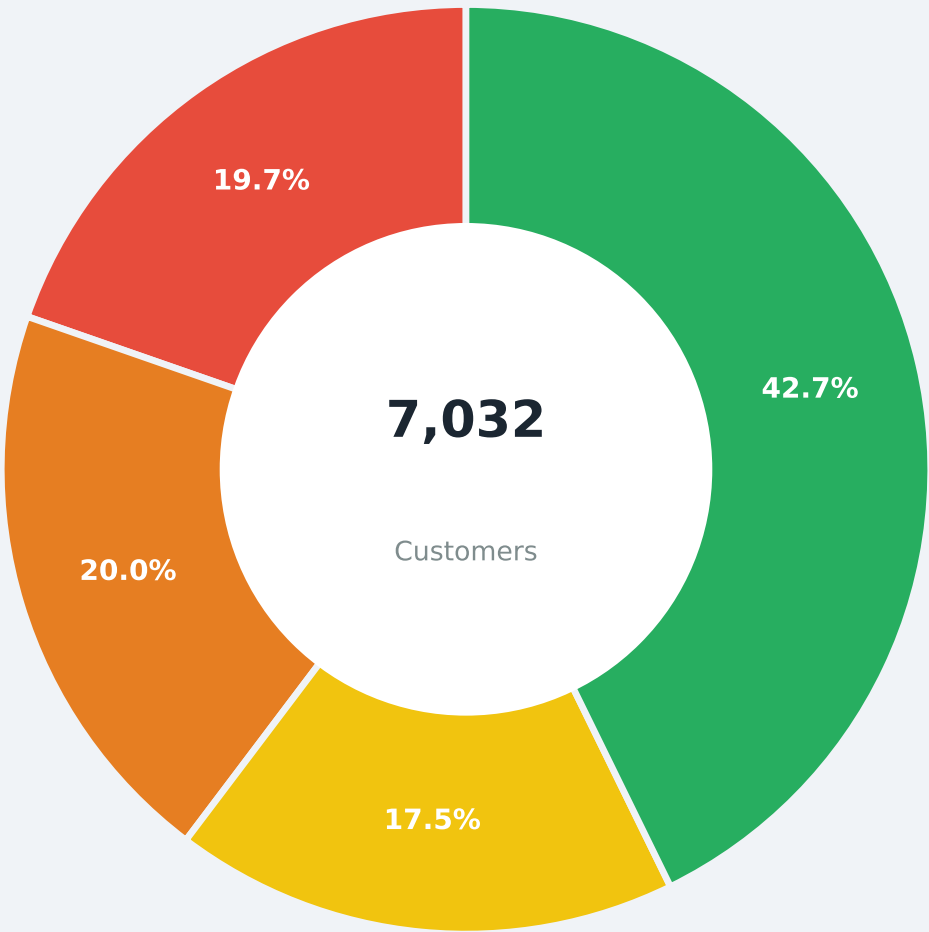
\$203K

Monthly Revenue at Risk

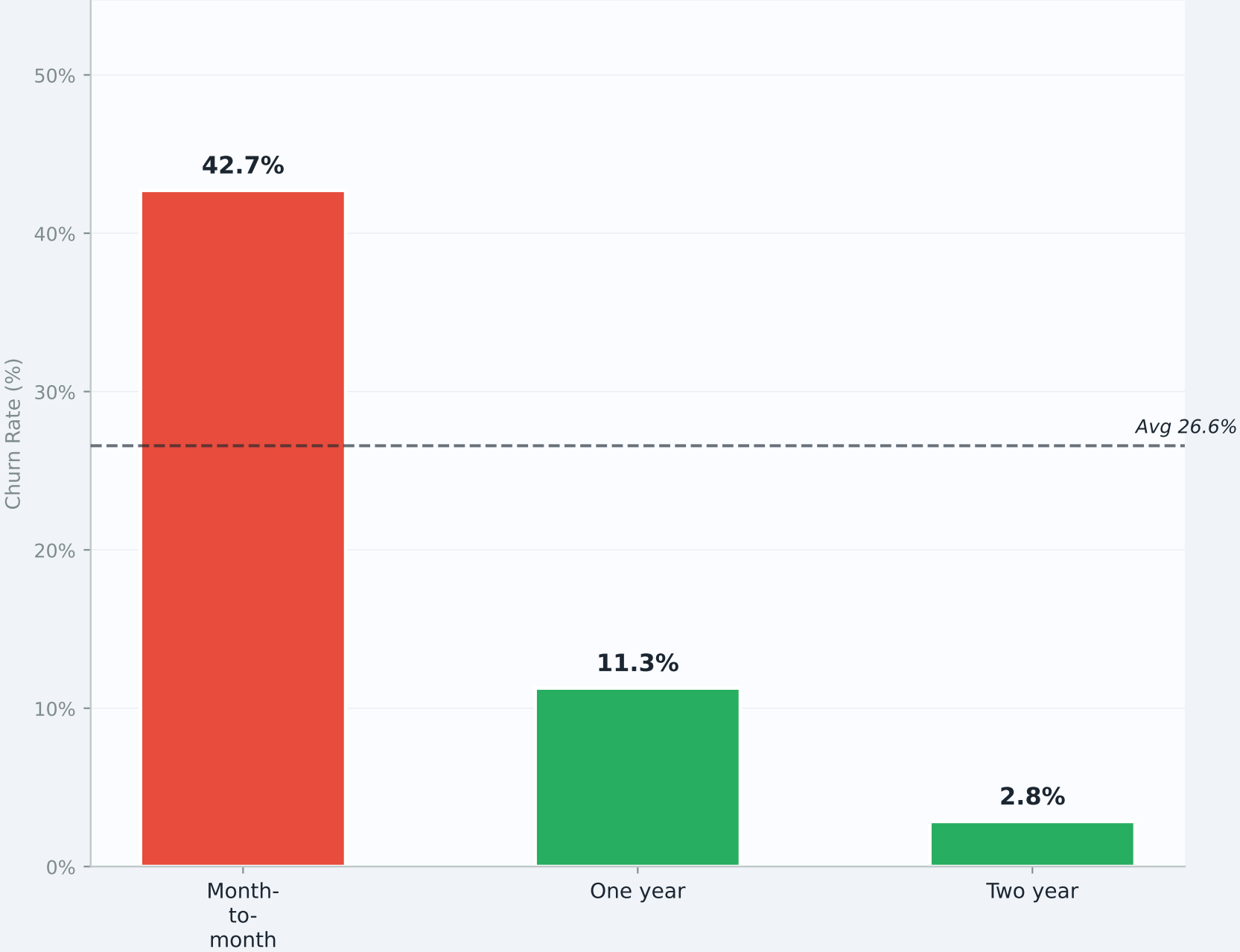
56 / 100

Portfolio Health Score

Customer Risk Tier Distribution



Churn Rate by Contract Type



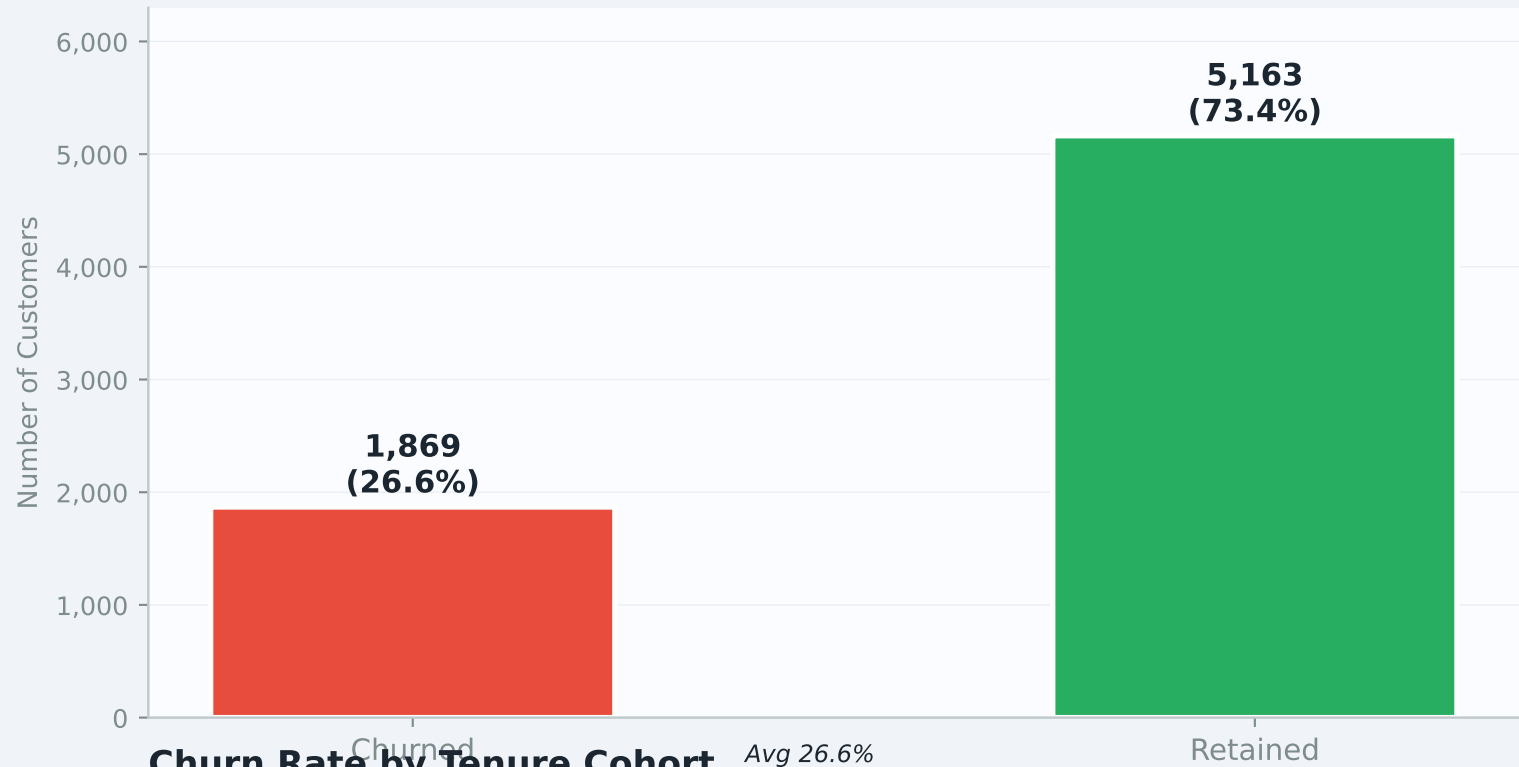
Key Insights

- Month-to-month customers churn at 42.7% — 1.6x the company average of 26.6%
- 1,383 customers are in the Critical risk tier — immediate retention action required
- Portfolio Health Score of 56/100 — \$203K in monthly revenue is at risk of loss

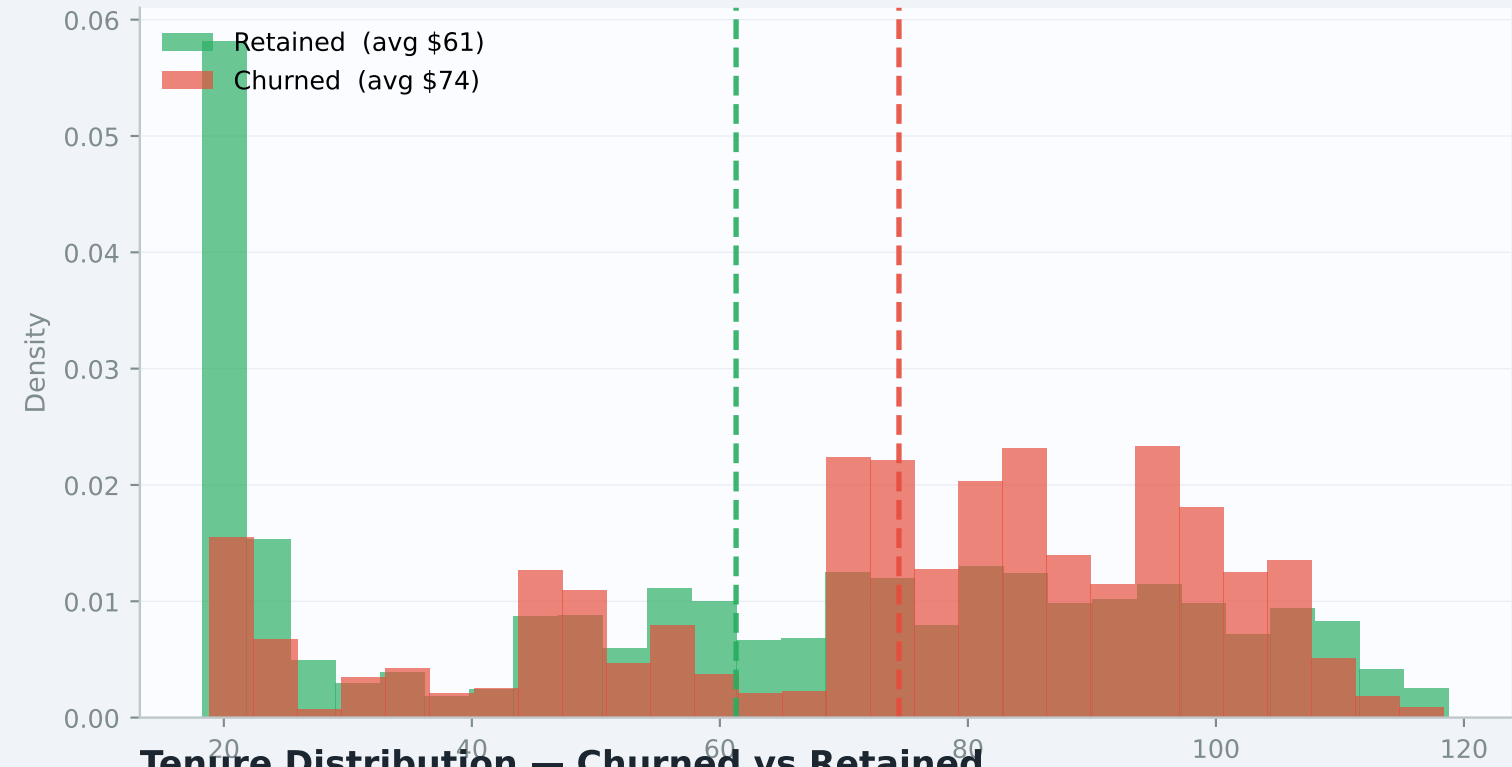
Churn Distribution Analysis

Who churns, when they churn, and what they pay — core behavioural patterns

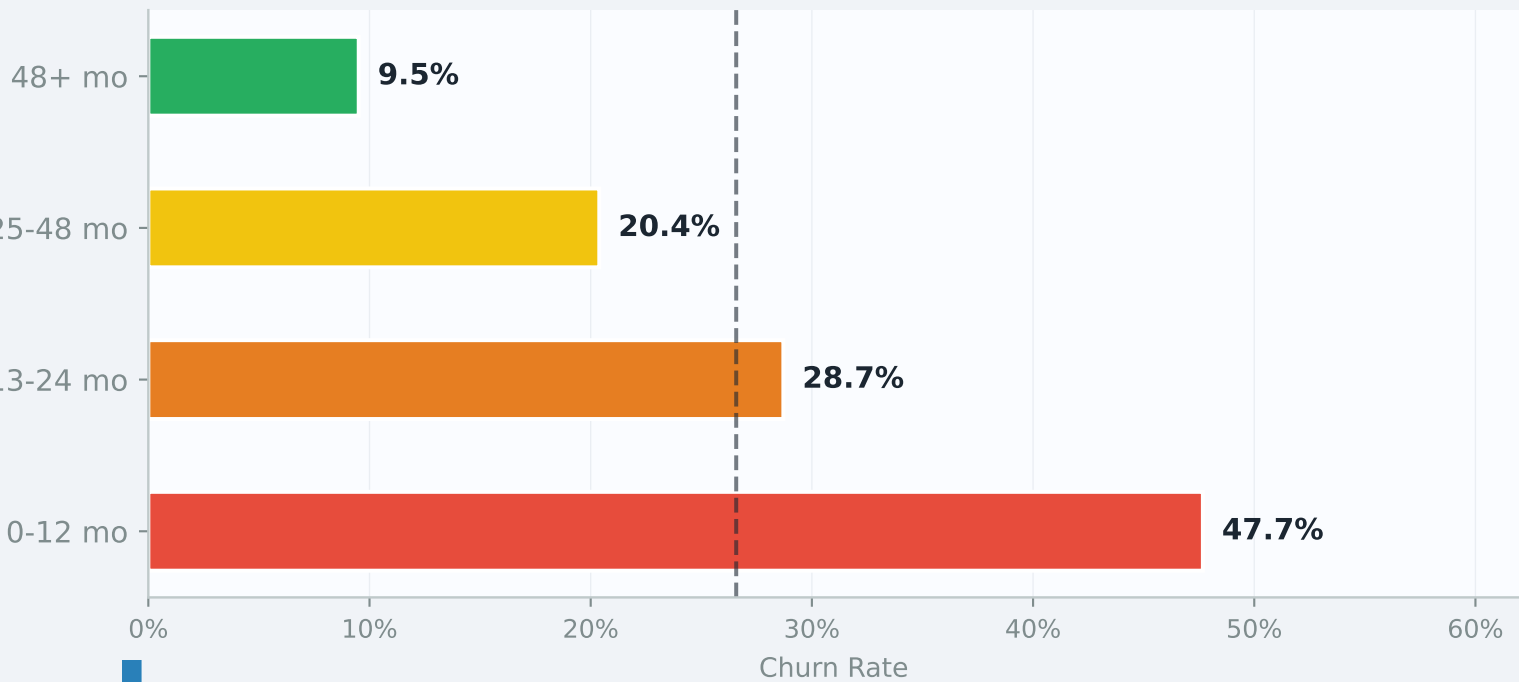
Churned vs Retained Customers



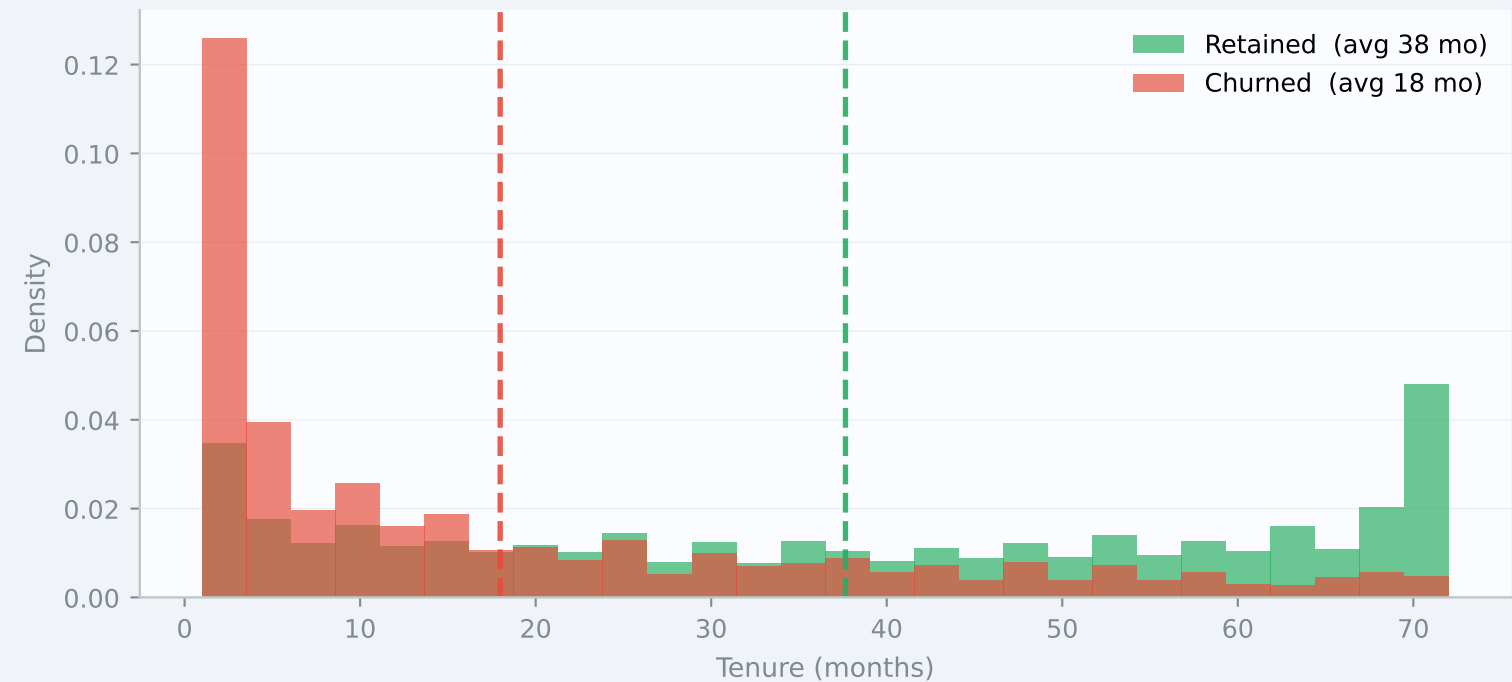
Monthly Charges — Churned vs Retained



Churn Rate by Tenure Cohort



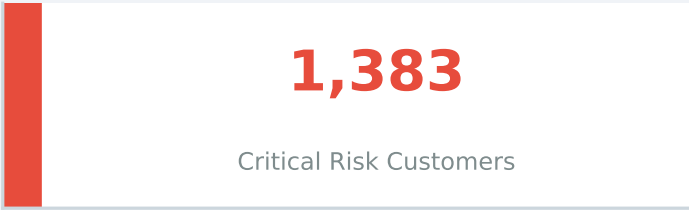
Tenure Distribution — Churned vs Retained



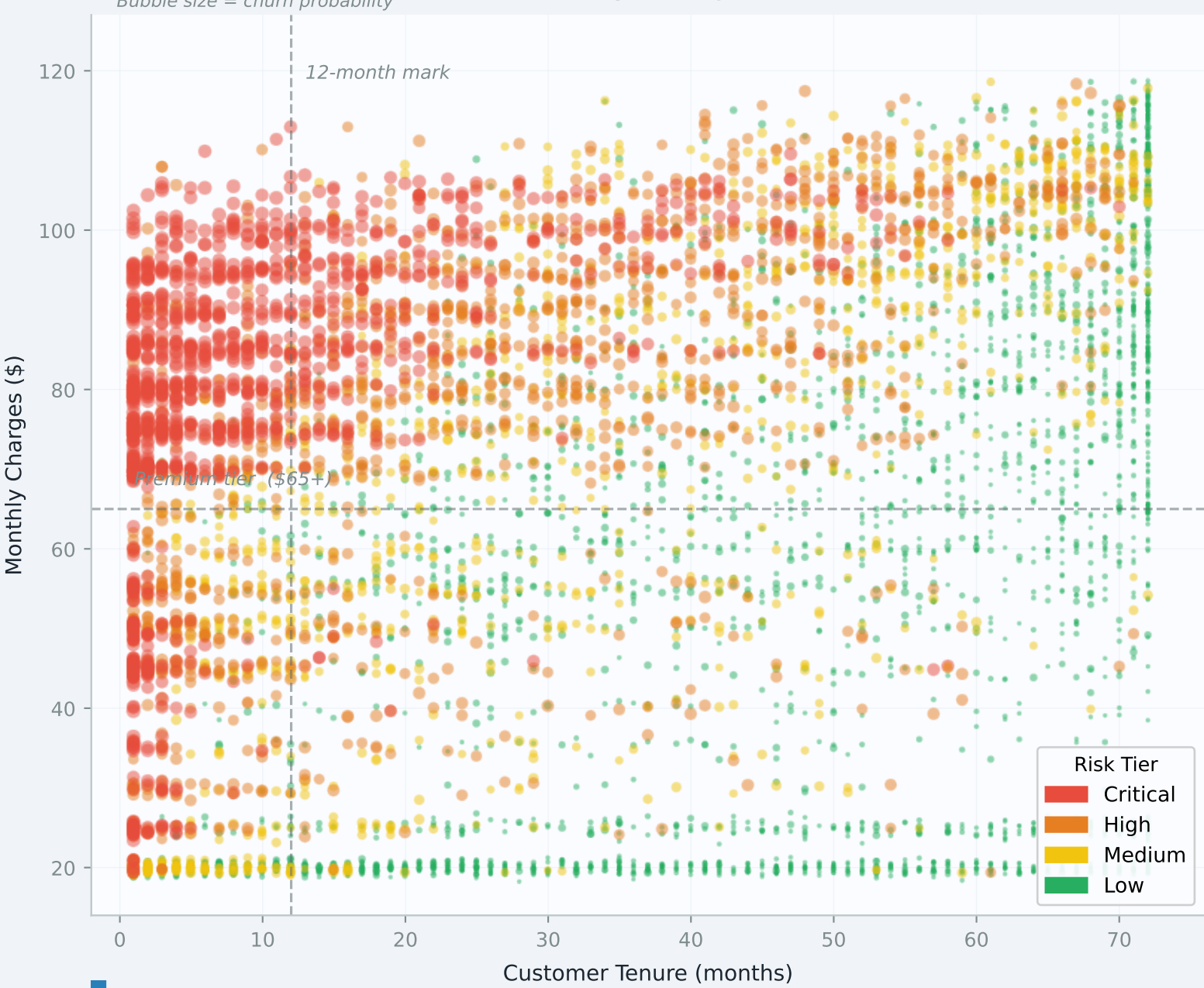
Key Insights

- New customers (0-12 mo) churn at 47.7% — 5.0× higher than loyal customers at 9.5%
- Churned customers pay 74/moavgvs61/mo for retained — higher charges = higher risk
- Average churned tenure is only 18 months vs 38 months for retained customers

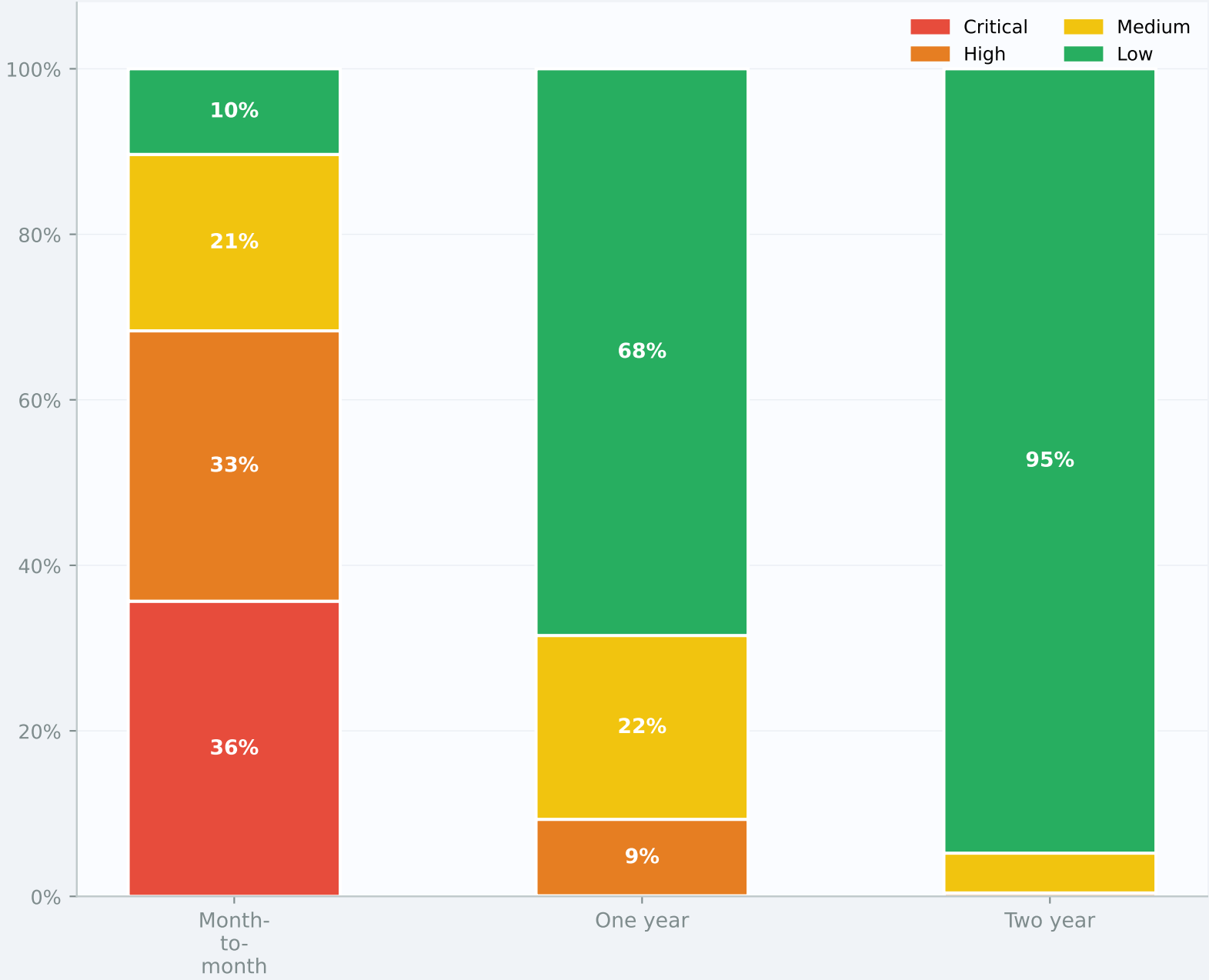
Which segments concentrate churn risk — scatter quadrants and risk-tier breakdowns



Risk Quadrant — Tenure vs Monthly Charges



Risk Concentration by Contract



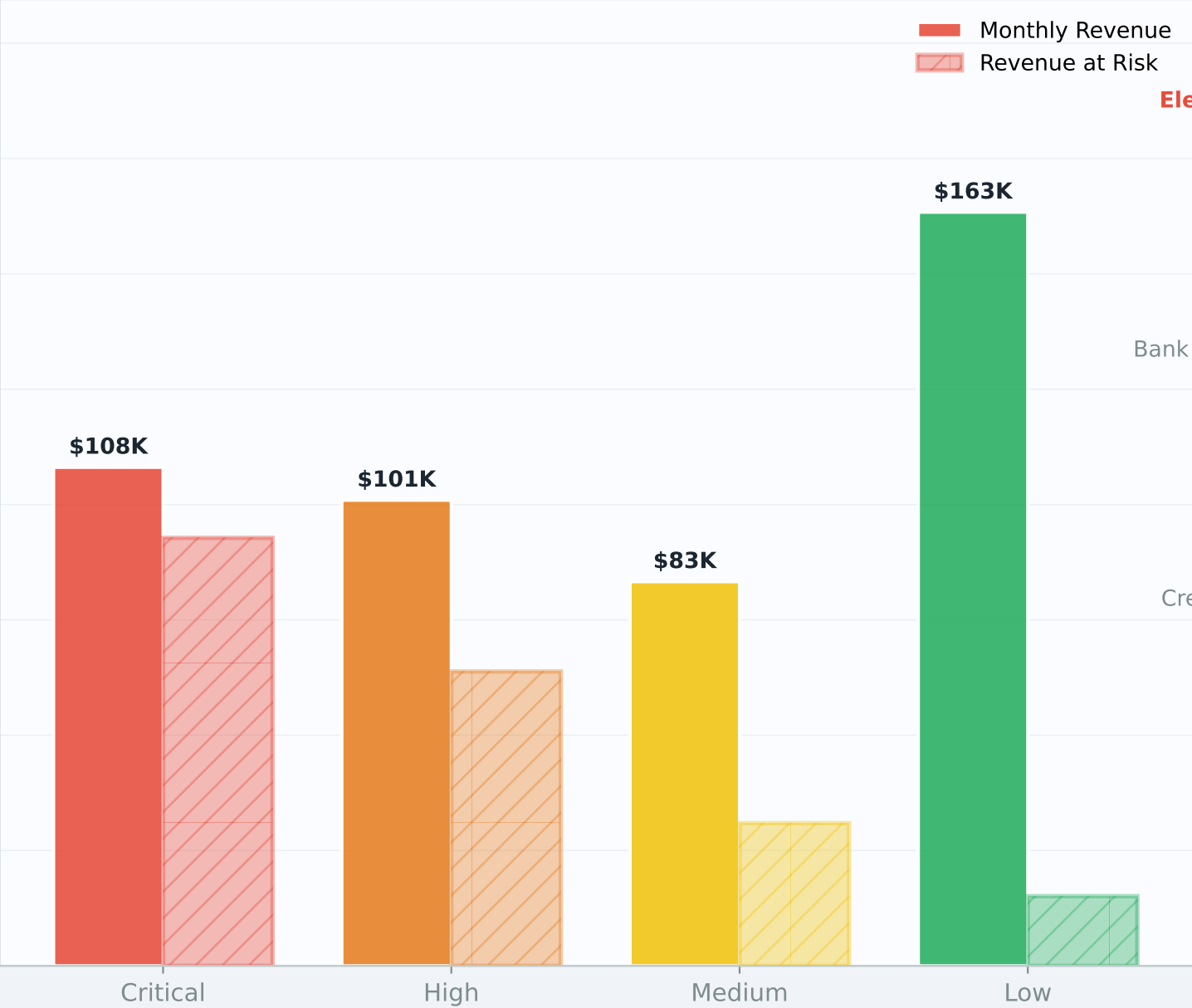
Key Insights

- Critical-tier customers are concentrated in month-to-month contracts — longest contracts have almost zero Critical-tier customers
- High tenure (48+ mo) and low monthly charges = Low risk quadrant — these loyal customers must be protected with proactive engagement
- Top-right quadrant (new + high charges) is the highest-priority retention zone — `{int((df["tenure"]<13)&(df["MonthlyCharges"]>65) & (df["RiskTier"]=="Critical")).sum() if False else crit_n:,}` Critical customers here

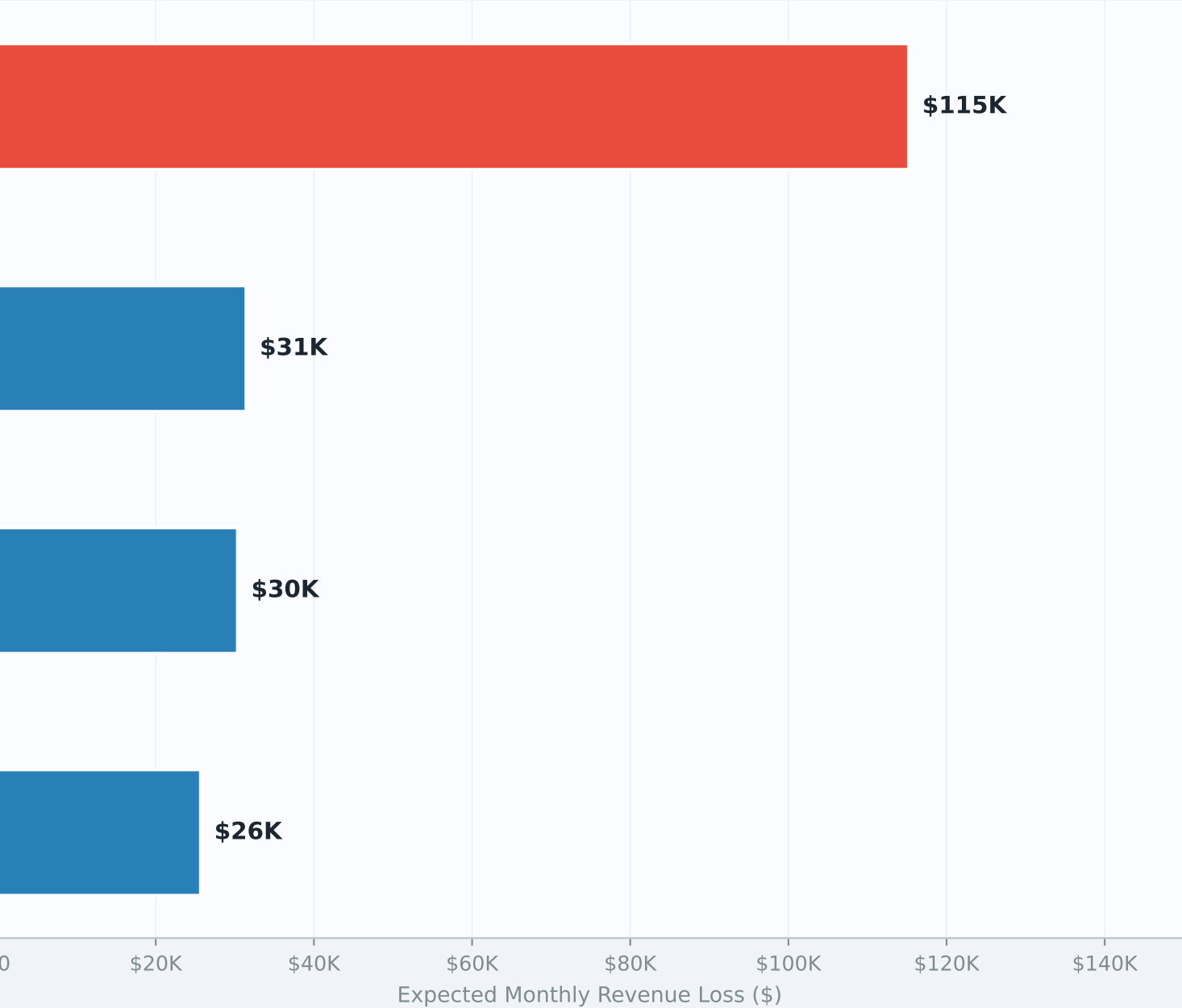
Financial exposure in dollar terms — what is at risk, where it sits, and who owns it



Monthly Revenue vs Revenue at Risk by Tier



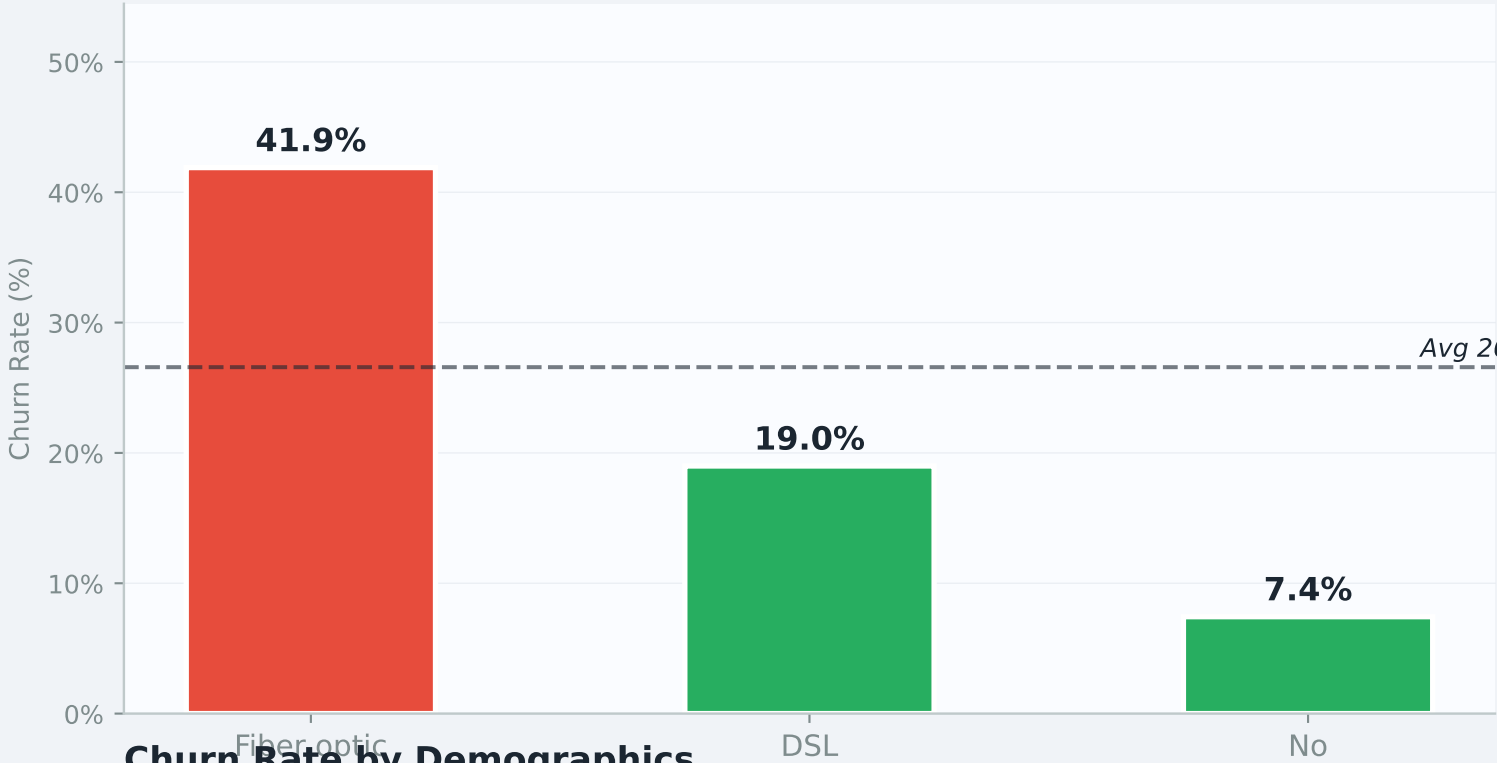
Expected Monthly Loss by Payment Method



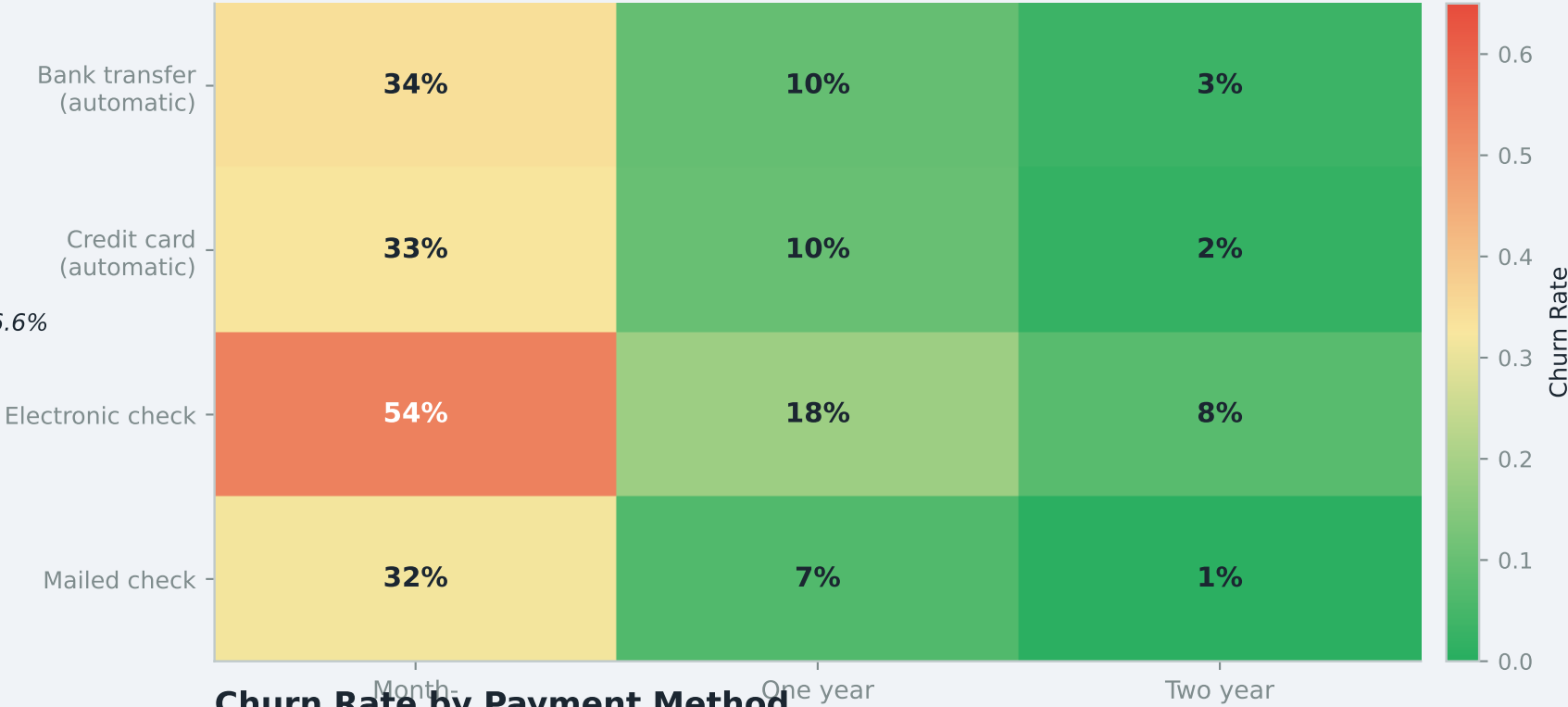
Key Insights

- Electronic check customers drive 57% of all expected revenue loss — \$115K/mo — auto-payment migration is a high-ROI retention lever
- Critical + High risk tiers hold 157K of 203K total monthly revenue at risk — focus there first
- Expected annual revenue loss of \$2.4M can be significantly reduced by addressing the top two risk tiers alone

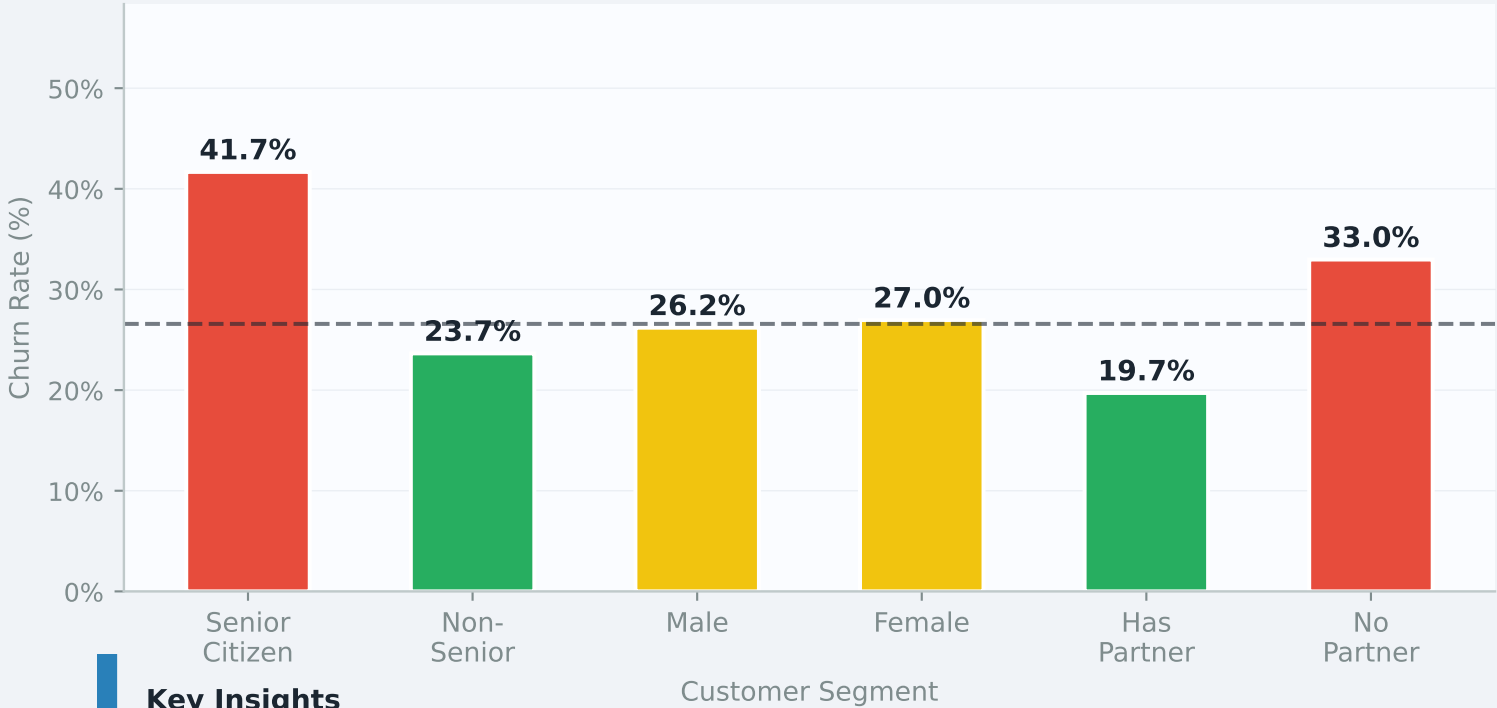
Churn Rate by Internet Service



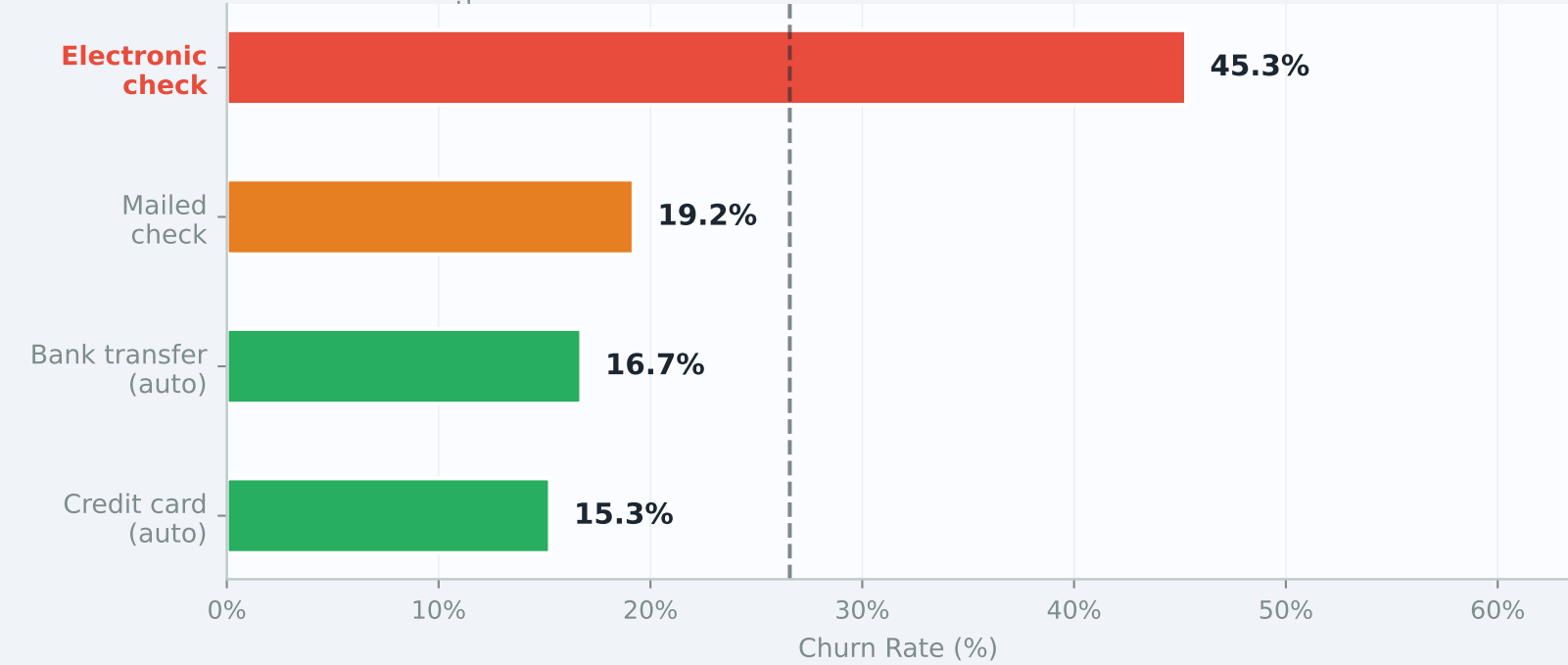
Churn Heat Map — Payment × Contract



Churn Rate by Demographics



Churn Rate by Payment Method

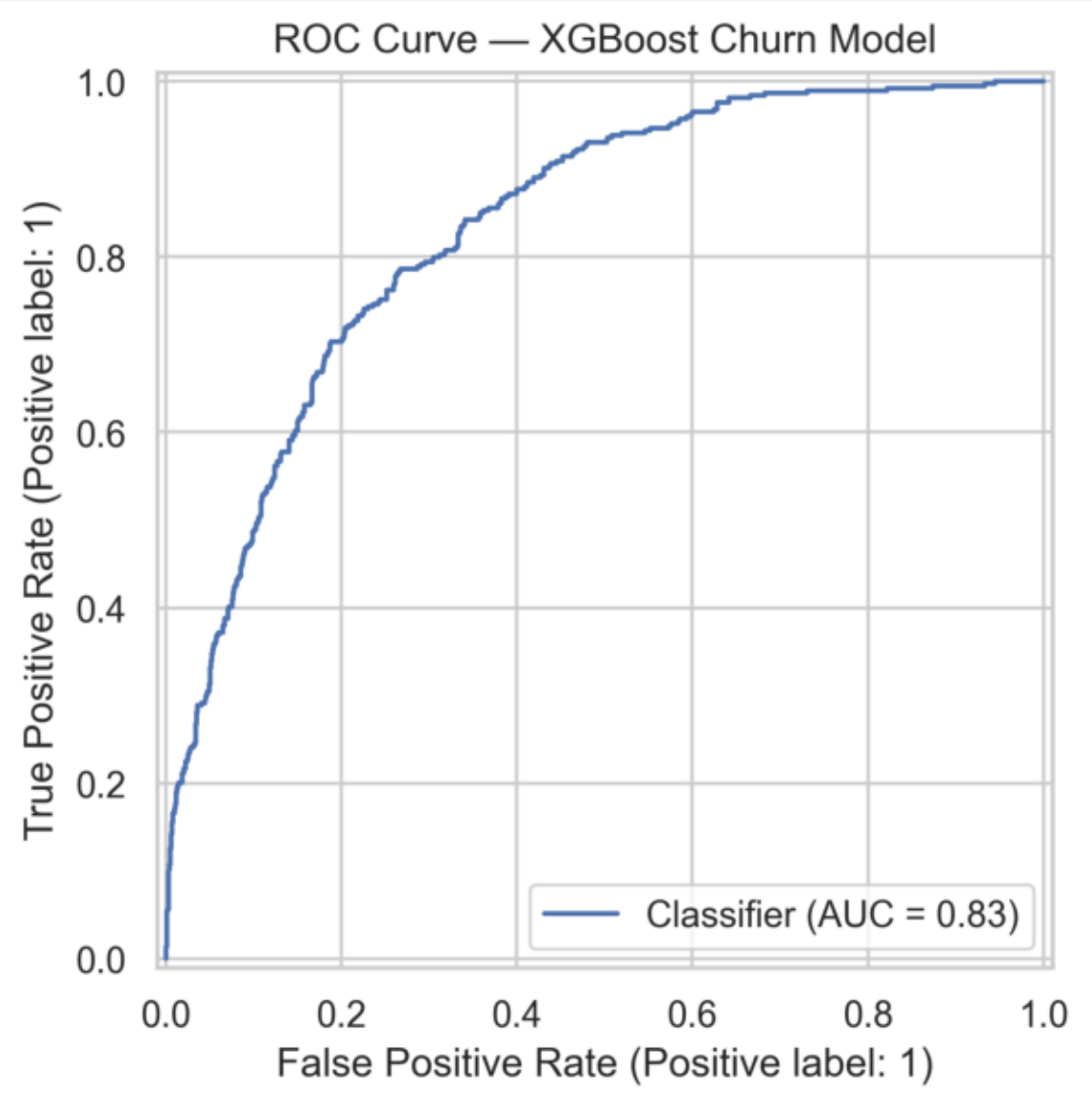
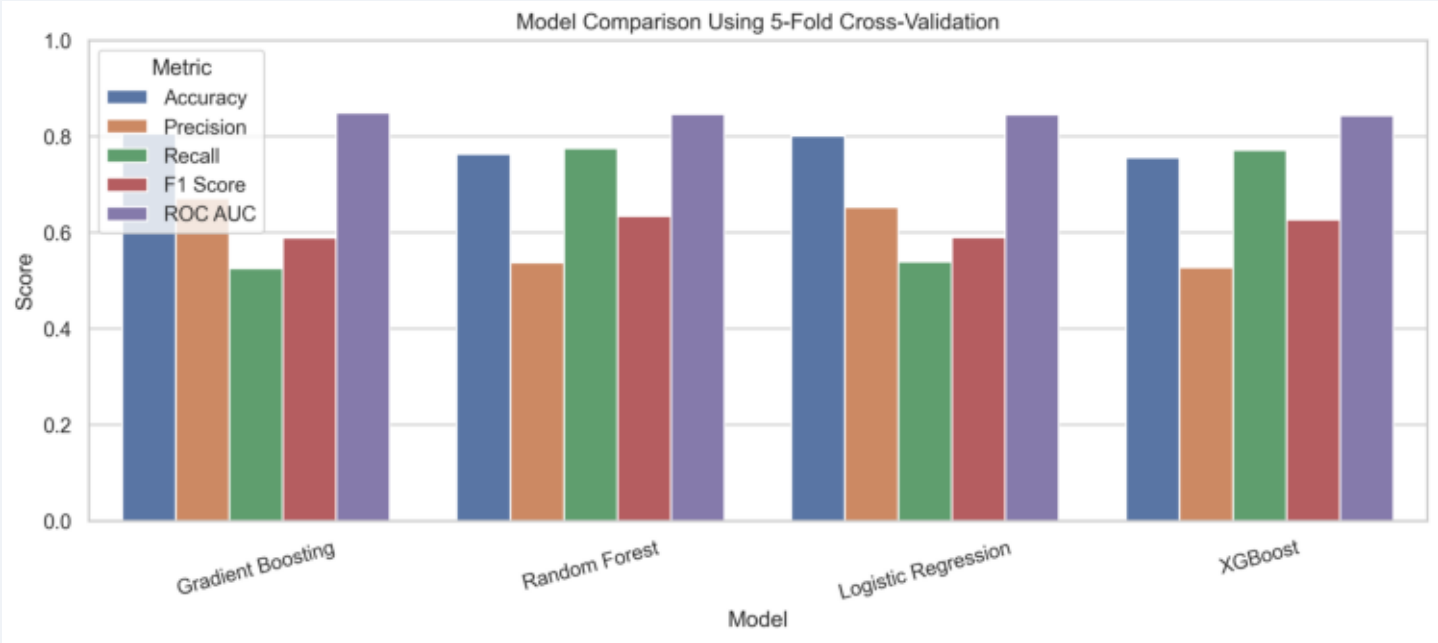
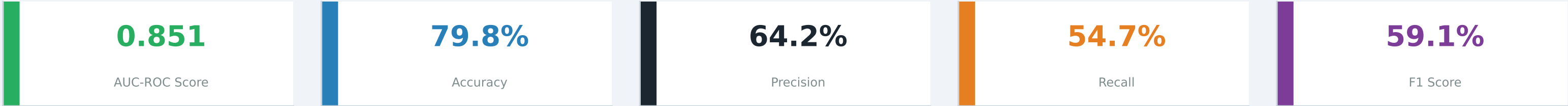


Key Insights

- Fiber optic customers churn at 41.9% — 1.6× the overall rate — the single riskiest internet segment
- Electronic check + Month-to-month = highest churn cell in the heat map — Electronic check customers churn at 45.3% vs 16.0% for auto-pay
- Customers with no dependents churn at 31.3% — family customers have more reasons to stay and churn far less

Predictive Model Performance

XGBoost churn classifier — accuracy, ROC-AUC, precision-recall and confusion matrix



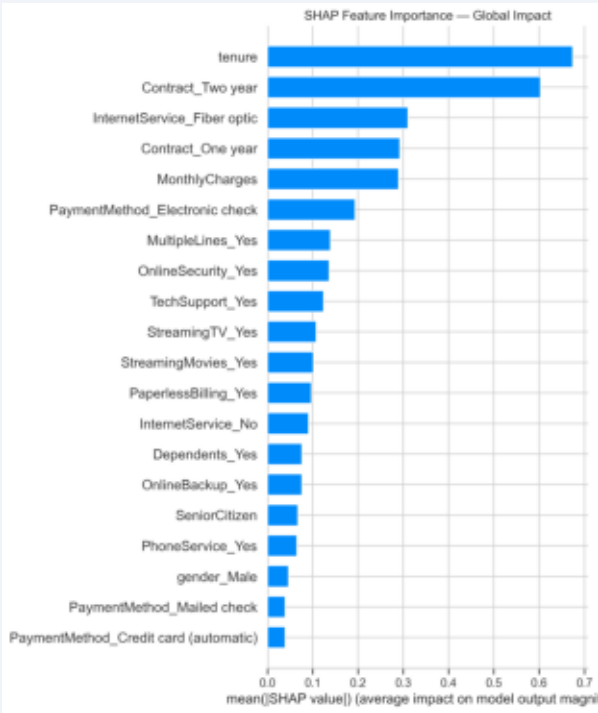
Key Insights

- AUC-ROC of 0.851 means the model correctly ranks a random churner above a random non-churner 85% of the time — well above random (0.50)
- Precision 64.2% / Recall 54.7% trade-off: tuned for recall to catch more churners at the cost of some false positives — correct for retention use cases
- XGBoost outperforms Logistic Regression and Random Forest on all metrics — selected as the production model for risk scoring

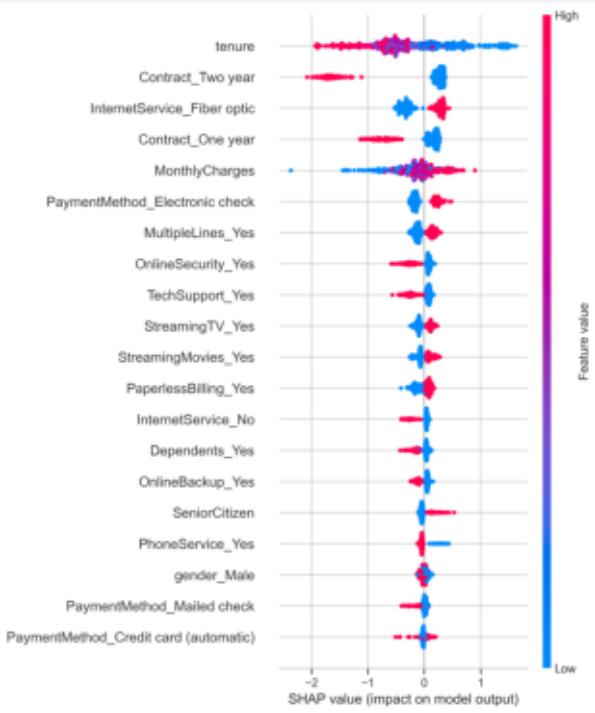
SHAP Feature Importance & Explainability

Which features drive churn predictions — global importance and individual-level impact

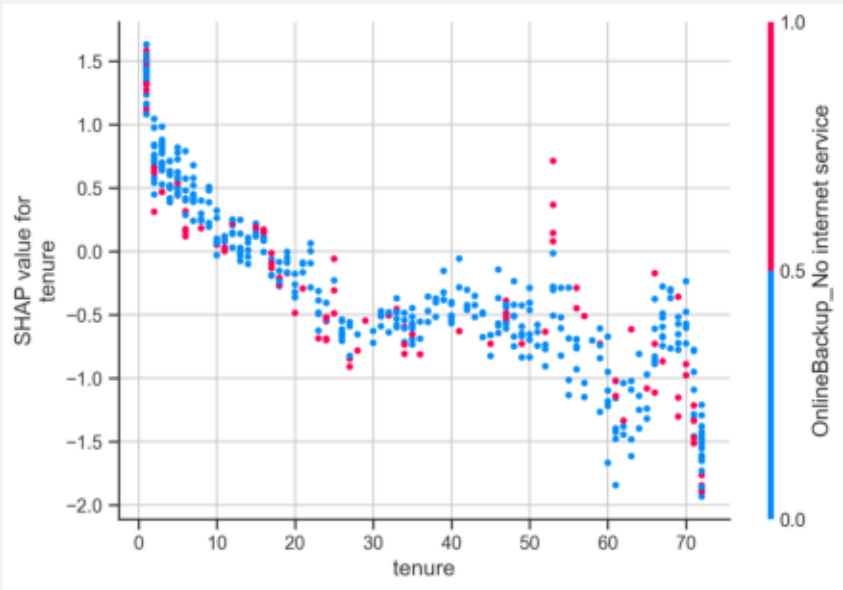
Mean |SHAP| Feature Importance



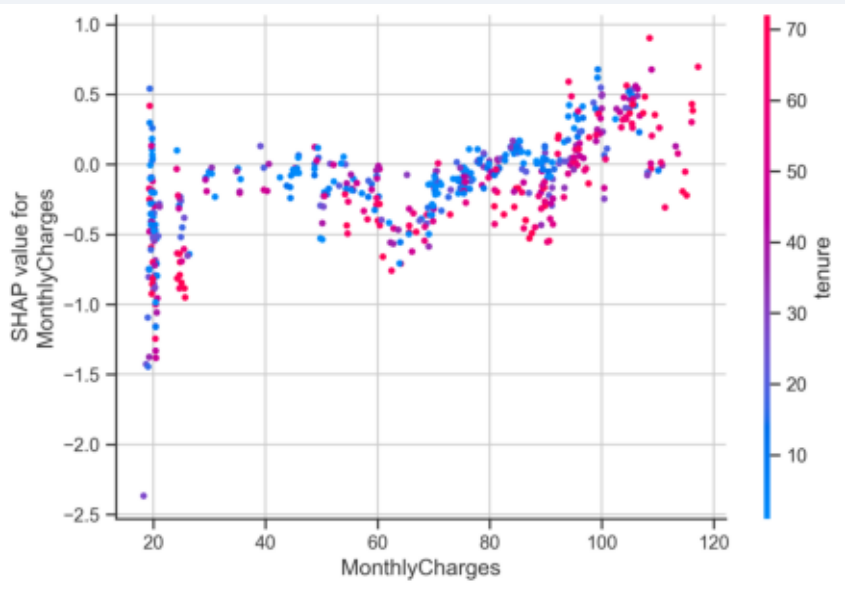
SHAP Beeswarm — Feature Impact Direction



SHAP Dependence — Tenure



SHAP Dependence — Monthly Charges



Key Insights

- Contract type is the #1 churn driver — month-to-month contract increases churn probability more than any other single feature
- Tenure has a strong negative SHAP effect — each additional month of tenure significantly reduces churn probability (loyalty compounds over time)
- Monthly charges above ~\$65 sharply increase churn risk — customers who feel overcharged leave; price anchoring helps retention

Retention War Room

Operational action centre — prioritised customer queue and revenue recovery potential

4,026

Customers Needing Action

57.3%

Action Rate

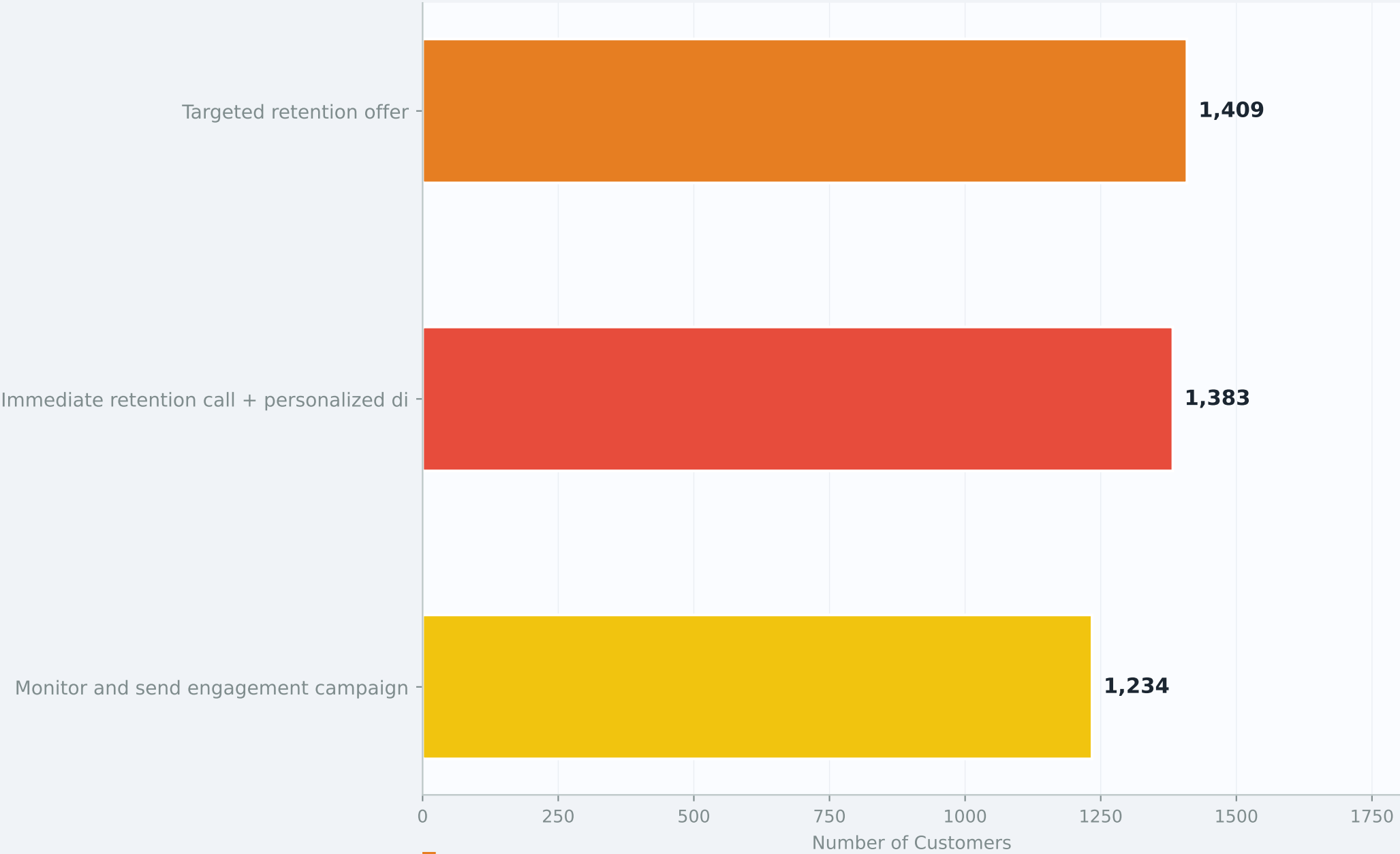
1,383

Critical Priority Customers

\$729K

Annual Recovery Potential

Customers by Recommended Retention Action



Priority Retention Queue (Top 15 by Churn Probability)

Customer ID	Risk	Churn Prob	Mo. Charges	Contract	Recommended Action
8149-RSOUN	Critical	98%	\$94	Month-to-month	Immediate retention call + personalized discount
4910-GMJOT	Critical	98%	\$95	Month-to-month	Immediate retention call + personalized discount
5178-LMXOP	Critical	98%	\$95	Month-to-month	Immediate retention call + personalized discount
9300-AGZNL	Critical	98%	\$94	Month-to-month	Immediate retention call + personalized discount
0107-YHINA	Critical	97%	\$100	Month-to-month	Immediate retention call + personalized discount
9497-QCMMS	Critical	97%	\$94	Month-to-month	Immediate retention call + personalized discount
5419-JPRRN	Critical	97%	\$101	Month-to-month	Immediate retention call + personalized discount
0295-PPHDO	Critical	97%	\$95	Month-to-month	Immediate retention call + personalized discount
7216-EWTRS	Critical	97%	\$101	Month-to-month	Immediate retention call + personalized discount
1069-XAIEM	Critical	97%	\$85	Month-to-month	Immediate retention call + personalized discount
5192-EBGOV	Critical	97%	\$86	Month-to-month	Immediate retention call + personalized discount
6861-XWTWQ	Critical	97%	\$99	Month-to-month	Immediate retention call + personalized discount
3178-FESZO	Critical	97%	\$100	Month-to-month	Immediate retention call + personalized discount
3068-OMWZA	Critical	97%	\$89	Month-to-month	Immediate retention call + personalized discount
9282-IZGQK	Critical	97%	\$85	Month-to-month	Immediate retention call + personalized discount

Key Insights

- 1,383 Critical customers need a personal retention call today — a 30% save rate on these alone recovers \$729K annually
- Electronic check + Month-to-month is the single highest-risk combination — offer auto-payment discounts to convert these customers first
- Top 15 queue customers all have >70% churn probability — personalised discount + contract upgrade is the recommended intervention